CS683 : Programming Assignment 2

Group Name: Cache Me If You Can

Group Members

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Task 1A: IP Stride Prefetcher

Overview

- The IP Stride prefetcher tracks memory accesses **per load instruction (IP)**.
- For each IP, it stores the last accessed cache line and the last stride.
- If the **same stride is seen twice in a row**, it prefetches the next 3 cache lines along that stride.

Data Structure

- A fully associative table of **24 entries**, each storing: `ip`, `last_cl_addr`, `last_stride` and an `lru` rank.
- Only the **lower 8 bits** of the IP are stored (`MASK_IP = 0xFF`).
- On a table miss, the least recently used entry is replaced.

Working

- The address is converted to a cache-line address (`addr >> 6`), so strides are measured in cache lines.
- **New IP:** an entry is allocated and no prefetch is issued.
- **Known IP:** the stride is computed as the difference between the current and last cache-line addresses (can be negative).
- If the stride is 0, the access is ignored.
- If the stride equals `last_stride`, lines $addr+s$, $addr+2s$ and $addr+3s$ are prefetched (`PREFETCH_DEGREE = 3`).
- Prefetching stops if the address crosses the current **4 KB page**.
- If the L1D MSHR is less than half full, the line is prefetched into **L1**; otherwise into **L2**.
- Finally, `last_cl_addr`, `last_stride` and the LRU rank are updated.

Limitations

- It can only detect a **constant stride**. For a repeating pattern like $1, 2, 3, 1, 2, 3, \ldots$, consecutive strides never match, so **no prefetches** are issued. This motivates the Stride Pattern Prefetcher (SPP).
- Using only 8 bits of the IP causes **aliasing** between different instructions.
- It has no confidence counter, and the prefetch degree is fixed.

Task 1B: Stride Pattern Prefetcher (SPP)

Design

- **IP Table (32 entries):** tracks each load IP's last address, last stride, a 2-bit uniform-stride confidence counter, and a 12-bit **signature** encoding its recent stride history.
- **Pattern Table (256 entries):** maps a signature to the next expected stride, with a 2-bit confidence counter.
- **Uniform first:** if the same stride repeats 3 times in a row, the IP is in uniform mode and prefetches $addr + k \cdot stride$.
- **Fallback to non-uniform:** when the stride breaks, the uniform confidence resets to 0 and prediction switches to the Pattern Table, walking ahead using each predicted stride to update the signature.
- **Prefetch filter (64 entries):** records recently prefetched lines, to drop duplicate prefetches and to count useful prefetches for the accuracy checker.
- All prefetches stay within the current 4 KB page.

Storage Overhead

Structure	Fields (bits)	Bits/entry	Entries	Total
IP Table	valid 1, IP tag 10, last addr 16, last stride 7, uniform conf 2, signature 12, LRU 5	53	32	1696
Pattern Table	next stride 7, confidence 2	9	256	2304
Prefetch filter	valid 1, used 1, tag 12	14	64	896
Throttling	issued 8, useful 9, degree 3	–	–	20
Total				**4916 bits ≈ 614.5 B**

Accuracy-Based Throttling

- Accuracy is measured over epochs of 256 issued prefetches: accuracy = useful/256.
- The degree for the next epoch is chosen as follows:

Accuracy	Degree
$\geq 75\%$	4
50–75%	3
25–50%	2
$< 25\%$	1

- **Why these thresholds:** equal 25% bands give one degree step per quarter of accuracy, which is simple to implement in hardware (comparisons against 64, 128 and 192 useful prefetches per epoch).

- **Why these degrees:** above 75% almost every prefetch is used, so prefetching further ahead hides more latency. Below 25%, most prefetches pollute the small L1D and waste MSHRs and bandwidth, so the degree drops to 1.
- **Why a maximum of 4:** with strides of 2–3 lines, 4 prefetches already reach 8–12 lines ahead; going further mostly runs into the page boundary or evicts useful L1D lines.
- **Epoch of 256:** long enough for a stable accuracy estimate, short enough to react to program phase changes.

Results

Trace	Config	IPC	Speedup	L1D MPKI	Acc. (%)	Cov. (%)
Trace1	No prefetch	0.2324	1.0000	10.68	–	–
	IP-stride	0.2334	1.0045	10.12	5.5	5.2
	SPP	0.2341	1.0075	9.74	12.8	8.8
Trace2	No prefetch	0.2503	1.0000	13.53	–	–
	IP-stride	0.2501	0.9992	13.43	2.9	0.7
	SPP	0.2506	1.0013	13.35	6.4	1.3
Trace3	No prefetch	0.3821	1.0000	3.40	–	–
	IP-stride	0.3829	1.0021	3.32	1.4	2.5
	SPP	0.3850	1.0075	3.11	4.9	8.6
Geomean speedup			IP-stride 1.0019, SPP 1.0054			

- Accuracy here is ChampSim's L1D measure (useful / issued), which only counts prefetched lines that reached the L1D before the demand access.
- Coverage = (baseline misses − prefetcher misses)/baseline misses.

SPP Internal Statistics

Trace	SPP accuracy	Non-uniform share	Epochs at degree 1/2/3/4 (%)
Trace1	69.1%	78.7%	0.1 / 14.4 / 48.1 / 37.4
Trace2	59.1%	88.6%	0.5 / 19.3 / 67.5 / 12.6
Trace3	59.4%	84.9%	0.7 / 13.7 / 77.0 / 8.6

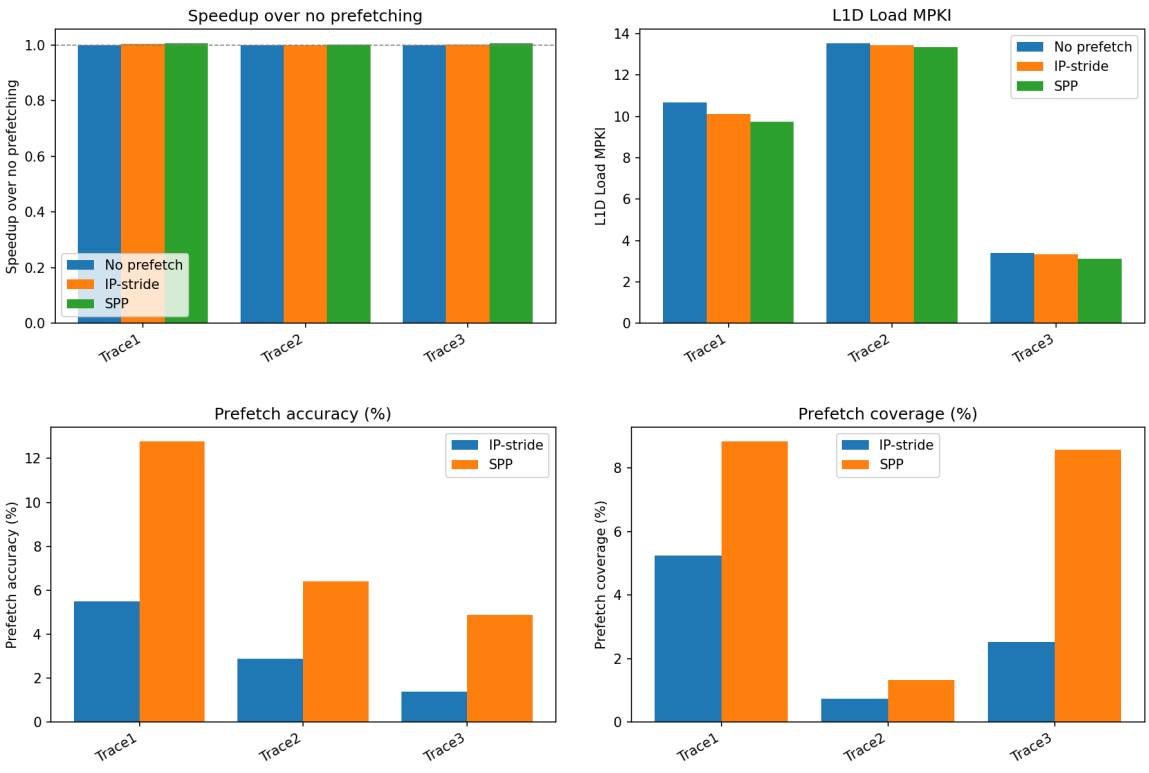


Figure 1: IPC improvement, L1D load MPKI, accuracy and coverage.